

Mother's Day Crossword

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1	2
3	

ONE ACROSS

The three ribbon-loop lengths on a Mother's Day wreath are positive integers a, b, c . They satisfy $a + b + c = 19$ and $ab + bc + ca = 99$. Find abc .

ONE DOWN

A digital Mother's Day card opens with a two-digit code equal to the last two digits of 12^{2026} . What is the code?

TWO DOWN

A Mother's Day ribbon is 8 inches long. It is tiled using 1-inch rose tiles and 2-inch heart tiles. No two heart tiles may touch. How many different tilings are possible?

THREE ACROSS

A Mother's Day gift tag is shaped like an isosceles trapezoid with bases 10 and 4. Given that its diagonals are perpendicular, find its area.